

Sales Analysis-Lower Manhattan Coffee Shop, January 1, 2023

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Data understanding

I used a coffee shop dataset with stores in three NYC neighborhoods, covering data from Jan-Jun 2023. I selected only the Lower Manhattan store for 1 Jan. This left 176 transactions. I grouped the products into drinks (coffee, tea, hot chocolate) and pastries (all bakery items). There were 143 drinks sold at \$3.0 on average, and 33 pastries sold at \$3.5 on average. I calculated these average prices manually from the unit price column and stored them in Python variables.

Results

Revenue from drinks was \$429.0 ($143 * \3.0) and from pastries was \$115.5 ($33 * \3.5). Total revenue was \$544.5. Drinks accounted for about 79% of total revenue, and pastries 21%. Revenue was \$44.5 over the \$500 target, but there is room to grow.

Recommendations

Drinks made up \$429.0 or 79% of the day's revenue. To maintain the momentum, the store should ensure quick drink service, perhaps with additional baristas during peak hours. However, this analysis is based on a single day's data and does not establish a clear trend. I would first analyze a week's worth of sales before implementing changes.

Pastries accounted for 21% of total revenue, with 33 sold. The store could offer bundle deals (e.g., coffee + croissant) to increase pastry sales. However, this analysis does not demonstrate that promotions will succeed, so we would need to observe sales during a promotional period.

Total revenue exceeded the \$500 target by just \$44.5. Management should monitor daily sales and identify low-revenue periods. Furthermore, since 1 Jan is a public holiday, sales volume may differ from a typical weekday. We should compare this day to an ordinary weekday.

Open questions

I estimated the average price per item from the price column (\$3.0 for drinks and \$3.5 for pastries), and the true average (by weight) is slightly different. I wonder if this discrepancy affects my conclusions.

Additionally, 1 Jan is a public holiday, and customer flow may vary significantly compared to a standard workday. I wonder whether insights derived from a single day can be generalized to the entire year.